

Electronics in Power Sector: Challenges & Opportunities

Power transmission losses in India are among the highest in the world at around 4 to 4.5 per cent. Distribution losses, which ideally should be confined to a single digit, hover around 20-30 per cent. This gives ample scope to technology providers to come up with solutions that help counter these losses



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India currently suffers from a major shortage of electricity generation capacity, even though it is the world's fourth largest energy consumer after United States, China and Russia. The International Energy Agency estimates India needs an investment of at least \$135 billion to provide universal access of electricity to its population – reports Wikipedia.

The International Energy Agency expects India to add between 600 GW and 1200 GW of new power generation capacity before 2050. The technologies and fuel sources India adopts, as it adds this electricity generation capacity, may make significant impact to global resource usage and environ-

mental issues.

Efficiency in electricity generation, transmission and distribution and use of renewable energy sources like solar PV cells are important to achieve the goal of electricity for all.

Electronics in power sector

In power sector, energy efficiency in electricity generation depends on the machines used for generation of electric power. The key to efficiency in energy generation is to control auxiliary power consumption at a plant.

"Auxiliary power consumption at a conventional plant is assumed at around 10 per cent, while at a modern power plant the same is estimated at 8 to 8.5 per cent. With the application of variable-drive frequency technology,

auxiliary power consumption can be reduced by approximately two per cent directly. This will also help improve heat rate of boiler, thus helping overall fuel efficiency of the boiler," says N. Venu, president and head of Power Systems Division, ABB Ltd, India.

This variable-frequency drive technology is contemporary technology used as a standard measure across the globe.

However, as machines used for generation of electric power are synchronous machines, these don't have much to do with power electronics. Transmission and distribution systems, on the other hand, have a lot of power electronic applications. In India, we use the same technology as in the developed world but with fewer installations.

"The quality of electric power is determined by the flat frequency and sinusoidal voltage profile it can maintain. Any deviation from the set values will affect the performance of the devices connected to the system. Hence it is absolutely essential to maintain proper quality of power at the user end," explains Prof. (Mrs) Mini Shaji Thomas, member, IEEE Educational Activities Board, chair, Section Outreach Committee (SEOC), and professor and CPIO, Jamia Millia Islamia, New Delhi.

Power electronics devices

Power electronic devices are classified as half controlled devices (thyristors) that are turned on by a control signal and turned off by the power circuit, whereas fully controlled devices such as power MOSFETs, IGBTs and IGBTs are turned on and off by control signals.